

Problem 1: Electricity Bill Calculator

Statement: An electricity board charges based on units consumed:

- First 100 units → \$1 per unit
- Next 100 units (101–200) → \$1.5 per unit
- Above 200 units → \$2 per unit

Write pseudocode that takes the number of units consumed and calculates the total bill.

Input: units = 250

Output: "Total Bill: \$350"

Problem 2: BMI Calculator with Category

Statement: Write pseudocode that takes a person's weight (kg) and height (m), calculates their BMI, then classifies them:

- BMI < 18.5 → "Underweight"
- 18.5 to 24.9 → "Normal"
- 25 to 29.9 → "Overweight"
- 30 and above → "Obese"

Input: weight = 70, height = 1.75

Output:

BMI: 22.86

Category: Normal

Problem 3: Triangle Type Checker

Statement: Write pseudocode that takes three sides of a triangle and determines its type:

- If any side is $\geq$ sum of the other two $\rightarrow$ "Not a valid triangle"
- If all three sides are equal $\rightarrow$ "Equilateral"
- If exactly two sides are equal $\rightarrow$ "Isosceles"
- Otherwise $\rightarrow$ "Scalene"

Input: side1 = 5, side2 = 5, side3 = 8

Output: "Isosceles"

Problem 4: Income Tax Calculator

Statement: A country's tax rule is:

- Income up to \$10,000 $\rightarrow$ No tax
- \$10,001 to \$30,000 $\rightarrow$ 10% tax on the amount above \$10,000
- Above \$30,000 $\rightarrow$ \$2,000 flat + 20% tax on the amount above \$30,000

Write pseudocode that takes a person's income and calculates the tax payable.

Input: income = 45000

Output: "Tax: \$5000"